

● HARD

● INFORMATION AND IDEAS

Mixed

10 questions · ~15 min

02

Q1

INFORMATION AND IDEAS

Black beans (*Phaseolus vulgaris*) are a nutritionally dense food, but they are difficult to digest in part because of their high levels of soluble fiber and compounds like raffinose. They also contain antinutrients like tannins and trypsin inhibitors, which interfere with the body's ability to extract nutrients from foods. In a research article, Marisela Granito and Glenda Álvarez from Simón Bolívar University in Venezuela claim that inducing fermentation of black beans using lactic acid bacteria improves the digestibility of the beans and makes them more nutritious.

Which finding from Granito and Álvarez's research, if true, would most directly support their claim?

- A. When cooked, fermented beans contained significantly more trypsin inhibitors and tannins but significantly less soluble fiber and raffinose than nonfermented beans.
- B. Fermented beans contained significantly less soluble fiber and raffinose than nonfermented beans, and when cooked, the fermented beans also displayed a significant reduction in trypsin inhibitors and tannins.
- C. When the fermented beans were analyzed, they were found to contain two microorganisms, *Lactobacillus casei* and *Lactobacillus plantarum*, that are theorized to increase the amount of nitrogen absorbed by the gut after eating beans.
- D. Both fermented and nonfermented black beans contained significantly fewer trypsin inhibitors and tannins after being cooked at high pressure.

ALSOL is a microcredit program in Mexico that makes small loans to female entrepreneurs who lack the collateral and credit history to secure financing from conventional banks. Borrowers use their business proceeds to repay loans in equal weekly installments and incur no penalty for missed payments other than lack of access to larger loans. Economists Gustavo Barboza and Sandra Trejos analyzed ALSOL data and found that rural borrowers, who mostly make and sell handicrafts, miss payments more often than urban borrowers do, partly because they sell their goods less frequently than they could. Barboza and Trejos claim that this behavior reflects strategic decisions that enable rural women to increase their profits per unit sold.

Which finding, if true, would most directly support Barboza and Trejos's claim?

- A. Many marketplaces require entrepreneurs to pay marketplace operators a fixed percentage of each day's proceeds in exchange for permission to sell goods there.
- B. Rural entrepreneurs can typically sell their goods for higher prices in cities than in their home areas, but the number of people selling competing goods tends to be higher in cities.
- C. Due to the lower costs they incur, rural entrepreneurs tend to require smaller initial loans than urban entrepreneurs do.
- D. The cost to rural entrepreneurs to bring their goods to towns with marketplaces is high but largely independent of the number of goods they bring.

When digging for clams, their primary food, sea otters damage the roots of eelgrass plants growing on the seafloor. Near Vancouver Island in Canada, the otter population is large and well established, yet the eelgrass meadows are healthier than those found elsewhere off Canada's coast. To explain this, conservation scientist Erin Foster and colleagues compared the Vancouver Island meadows to meadows where otters are absent or were reintroduced only recently. Finding that the Vancouver Island meadows have a more diverse gene pool than the others do, Foster hypothesized that damage to eelgrass roots increases the plant's rate of sexual reproduction; this, in turn, boosts genetic diversity, which benefits the meadows' health overall.

Which finding, if true, would most directly undermine Foster's hypothesis?

- A. At some sites in the study, eelgrass meadows are found near otter populations that are small and have only recently been reintroduced.
- B. At several sites not included in the study, there are large, well-established sea otter populations but no eelgrass meadows.
- C. At several sites not included in the study, eelgrass meadows' health correlates negatively with the length of residence and size of otter populations.
- D. At some sites in the study, the health of plants unrelated to eelgrass correlates negatively with the length of residence and size of otter populations.

Electronic music pioneer Wendy Carlos is credited with the music for three feature films: *A Clockwork Orange* (1971), *The Shining* (1980), and *Tron* (1982). However, her musical score for *A Clockwork Orange* is mostly made up of her arrangements of Ludwig van Beethoven's work. Also, almost all the music that she and Rachel Elkind composed for *The Shining* was unused by director Stanley Kubrick. It did not appear in the film. Of the three films, *Tron* is the one in which audiences can hear the most of Carlos's original compositions.

Which choice best states the main idea of the text?

- A. Of the three films for which Carlos is credited, *Tron* features the most original music from her.
- B. The director of *The Shining* used most of the music that Carlos composed for it.
- C. Beethoven is widely considered to be a more important composer than Carlos.
- D. Carlos is a notable innovator among film composers in the 1970s and 1980s.

In Asiya Wadud’s 2022 poem “Shorn, Treaded Red,” the poet contemplates a painting that has inspired her: Etel Adnan’s 2020 work *Satellites 27*. The painting, which features overlapping geometric shapes, fuels the poem’s exploration of temporality and identity. _____ blank in responding to Adnan’s artwork, Wadud’s poem reflects on the relationship between poetry and other art forms.

Which choice completes the text with the most logical transition?

- A. In other words,
- B. For instance,
- C. What’s more,
- D. Conversely,

While researching a topic, a student has taken the following notes:

- The factors that affect clutch size (the number of eggs laid at one time) have been well studied in birds but not in lizards.
- A team led by Shai Meiri of Tel Aviv University investigated which factors influence lizard clutch size.
- Meiri's team obtained clutch-size and habitat data for over 3,900 lizard species and analyzed the data with statistical models.
- Larger clutch size was associated with environments in higher latitudes that have more seasonal change.
- Lizards in higher-latitude environments may lay larger clutches to take advantage of shorter windows of favorable conditions.

The student wants to emphasize the aim of the research study. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. Researchers wanted to know which factors influence lizard egg clutch size because such factors have been well studied in birds but not in lizards.
- B. After they obtained data for over 3,900 lizard species, researchers determined that larger clutch size was associated with environments in higher latitudes that have more seasonal change.
- C. We now know that lizards in higher-latitude environments may lay larger clutches to take advantage of shorter windows of favorable conditions.
- D. Researchers obtained clutch-size and habitat data for over 3,900 lizard species and analyzed the data with statistical models.

In 2018, a team of researchers led by Dr. Caitlin Whalen compiled every available measurement of ocean mixing rates from the past two decades. With this novel data set, the team was able to determine how current-driven mixing varies across _____ blank and what impact it has on the distribution of heat and nutrients in the ocean.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. regions,
- B. regions:
- C. regions;
- D. regions

Bengali author Toru Dutt's *A Sheaf Gleaned in French Fields* (1876), a volume of English translations of French poems, _____ blank scholars' understanding of the transnational and multilingual contexts in which Dutt lived and worked.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. has enhanced
- B. are enhancing
- C. have enhanced
- D. enhance

Text 1

Chemically analyzing bulk rock-core samples from Australia's Mount McRae Shale, Ariel Anbar et al. found a transient but significant increase in molybdenum (Mo) at a point corresponding to roughly 2.5 billion years ago (Ga). On Earth, Mo is released mainly through oxidative weathering of minerals; Anbar et al. therefore concluded that atmospheric oxygen briefly increased 2.5 Ga, then returned to its earlier negligible level.

Text 2

Sarah Slotznick et al. reexamined the Mount McRae Shale core. Since chemically analyzing bulk samples can occlude contextual details, Slotznick et al. also employed high-resolution microscopy, which revealed volcanic debris—a known Mo host—around 2.5 Ga and microfractures in the surrounding matrix. The researchers assert that fluid could have reached the debris through the microfractures and initiated oxidative weathering long after debris deposition.

Based on the texts, Anbar et al. (Text 1) and Slotznick et al. (Text 2) would most likely disagree about the answer to which question about the portion of the Mount McRae Shale rock-core corresponding to 2.5 Ga?

- A. Does the increase in Mo in that portion suggest that atmospheric oxygen levels were not negligible before 2.5 Ga?
- B. Is the increase in Mo in that portion indicative of an increase in atmospheric oxygen dating to the same time?
- C. Is the increase in Mo in that portion attributable to oxidative weathering of the material found in that portion?

- D. Did chemically analyzing bulk samples lead to a false impression that there is an increase in Mo in that portion?

K.D. Leka and colleagues found that the Sun’s corona provides an advance indication of solar flares—intense eruptions of electromagnetic radiation that emanate from active regions in the Sun’s photosphere and can interfere with telecommunications on Earth. Preceding a flare, the corona temporarily exhibits increased brightness above the region where the flare is _____ blank.

Which choice completes the text with the most logical and precise word or phrase?

- A. antecedent
- B. impending
- C. innocuous
- D. perpetual